A Compact, Low Input Capacitance Neural Recording Amplifier

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***Abstract*—Conventional capacitively coupled neural recording amplifiers often present a large input load capacitance to the neural signal source and hence take up large circuit area. They suffer due to the unavoidable trade-off between the input capacitance and chip area versus the amplifier gain. In this work, this trade-off is relaxed by replacing the single feedback capacitor with a clamped T-capacitor network. With this simple modification, the proposed amplifier can achieve the same mid-band gain with less input capacitance, resulting in a higher input impedance and a smaller silicon area. Prototype neural recording amplifiers based on this proposal were fabricated in 0.35 μm CMOS, and their performance is reported. The amplifiers occupy smaller area and have lower input loading capacitance compared to conventional neural amplifiers. One of the proposed amplifiers occupies merely 0.056 mm^2. It achieves 38.1-dB mid-band gain with 1.6 pF input capacitance, and hence has an effective feedback capacitance of 20 fF. Consuming 6 μW, it has an input referred noise of 13.3 $\mu\mathrm{V}_{\mathrm{rms}}$ over 8.5 kHz bandwidth and NEF of 7.87. *In-vivo* recordings from animal experiments are also demonstrated.**

***Index Terms*—Biopotential amplifier, CMOS, input impedance, neural recording.**

I. Introduction

NEURAL signal acquisition is an important part of modern physiological research and its applications. Neural signal potentials recorded from electrodes are weak in amplitude and need to be pre-amplified prior to any signal processing. A preamplifier is commonly implemented, shown in Fig. 1, as a differencing capacitively coupled amplifier (DCCA) [1]. The preamplifier can also be implemented as a single-ended capacitively coupled amplifier (SCCA) [2]. Although suitable for bipolar neural signal recording, the capacitively coupled amplifier is being increasingly adopted as the front-end preamplifier for recording other biopotential signals [3]–[6]. The input AC-coupling capacitors C_{in} inherently reject DC offsets introduced by the electro-chemical interaction at the electrode-tissue site. Low noise efficiency factor (NEF) is easily achieved by optimizing the single operational transconductance amplifier element (OTA) in this circuit [7].

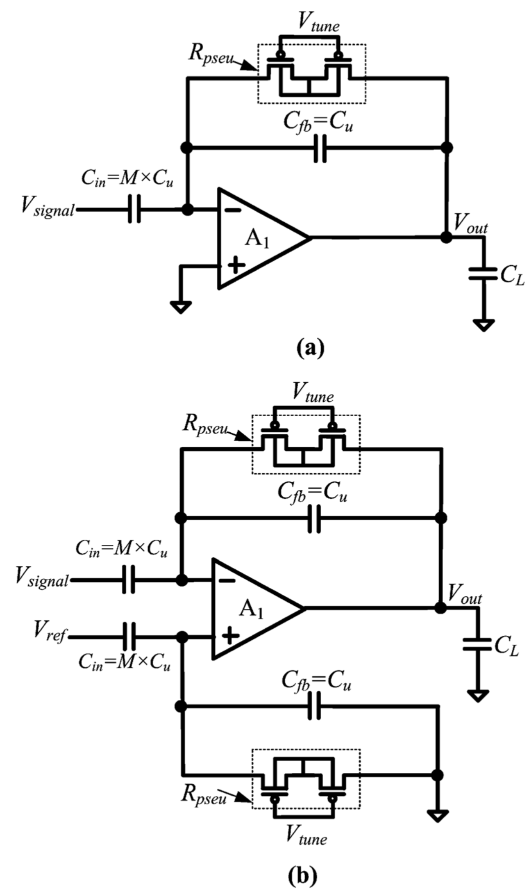

Fig. 1. (a) Schematic of the conventional SCCA [2]. (b) Schematic of the conventional DCCA [1].

For the conventional DCCA and SCCA, the mid-band gain as defined in (1) is determined by the ratio of input capacitor C_{in} to the feedback capacitor C_{fb}. When C_{in} is implemented with multiples of a unit capacitor C_u and C_{fb} is equal to C_u, the mid-band gain is equal to a factor M. The lower and upper 3-dB cutoff frequencies are also defined in (2) and (3), respectively. The total number of unit capacitors, $\mathrm{T_{CU}}$ needed to implement a fixed gain is defined in (4).

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{C_{in}}{C_{fb}} = M \tag{1}$$

$$f_L = \frac{1}{2\pi C_{fb} R_{pseu}} \tag{2}$$

$$f_U = \frac{g_m}{M C_L} \tag{3}$$

$$\mathrm{T_{CU}} = 2(M+1) \tag{4}$$

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To achieve low noise performance, the gain of the amplifier is usually large. Previously reported amplifiers [5]–[11] have achieved gains around 40 dB with an input capacitance ranging from 10 pF to 20 pF. From (1) and (4), we can see that input capacitance and hence area can be reduced, but at the expense of reduced gain. Most amplifiers occupy no less than 0.1 mm^2 of silicon area, of which a high percentage is occupied by the input capacitors. For applications that require a large number of recording channels or a smaller chip area, it is important to minimize the area of the capacitors within the recording amplifier.

The large input capacitance (10–20 pF) also translates to equivalent input impedance that, in connection with the electrode impedance, form a frequency-dependent potential divider circuit at the input of the amplifier. If the input impedance is insufficiently high, the recorded neural signal will not only be attenuated, but it will also undergo amplitude and phase distortion [12], [13]. In the case of electrodes that are used for chronic neural recording, there is also the problem of gradual onset of tissue fibrosis, which increases impedance between the neuron and electrode [14]–[16] and therefore leads to further signal attenuation and distortion. The effective common mode rejection ratio will also be degraded if the signal and reference electrode impedances are slightly mismatched [6]. Hence, there is a need to increase the input impedance of the neural recording amplifier.

As defined in (1), to obtain a large mid-band gain, either the input capacitance needs to be increased or the feedback capacitance lowered. The former will directly decrease the input impedance and increase the silicon area, while the latter is limited by the parasitic capacitance, mismatch, and the constraint of the process technology. On the other hand, to reduce the input capacitance (and area), the mid-band gain has to be sacrificially reduced if the feedback capacitor is constrained as mentioned before [11], [17]. Thus a trade-off is unavoidable. Such a trade-off can be characterized by the effective feedback capacitance, C_{fbeq}, and is equivalent to the input capacitance to gain ratio, C_{in}/Gain (fF.V/V), as first mentioned in [18]. A low effective feedback capacitance means that a smaller input capacitance, and hence less area is required to implement the given gain. State-of-the-art neural amplifiers have effective feedback capacitances no less than 100 fF. To reduce the chip area, it is possible to share the amplifier and the reference input capacitor amongst the channels [10]. However the common mode rejection ratio (CMRR) and inter-channel crosstalk increases with the number of shared channels and yet the effective feedback capacitance per amplifier remains high.

This paper proposes a neural amplifier circuit topology that relaxes the input capacitance (and area) versus gain constraint. The proposed topology can be used to either achieve a medium amplifier gain specification with much smaller input capacitance and silicon area or a higher mid-band gain without the penalty of increased input capacitance and silicon area. The concept of the proposed topology was first reported in [18] and will be further described with additional work in Section II. To demonstrate the feasibility of this proposed topology, some neural recording amplifiers are implemented, and the detailed circuit designs discussed in Section III. Section IV presents measurement results of the amplifiers together with some *in-vivo* experimental results.

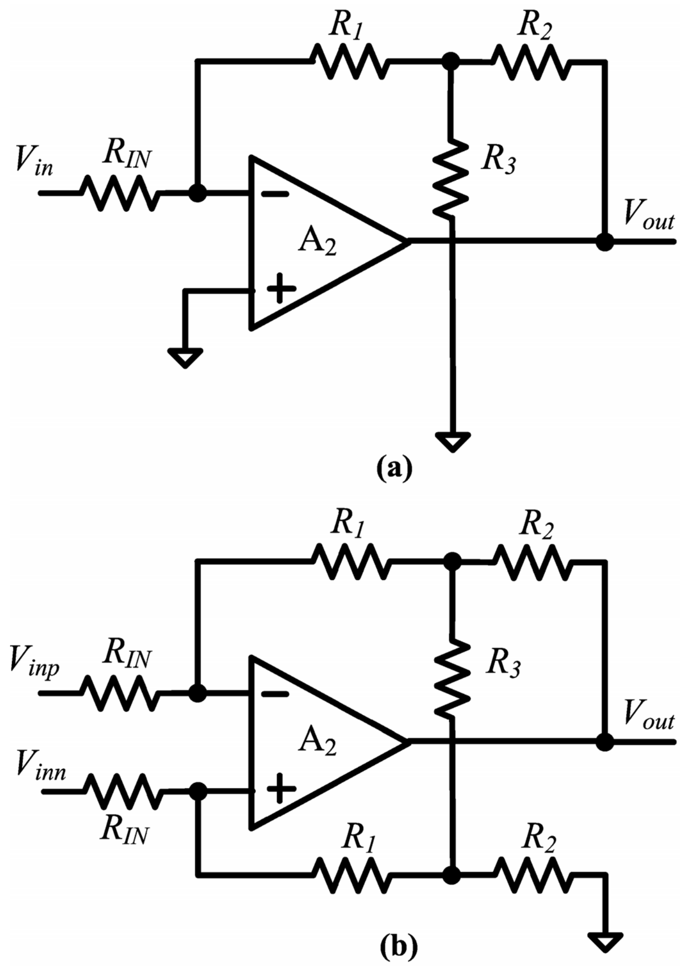

Fig. 2. Schematics of (a) a single-ended amplifier with T-resistor feedback path and (b) a differencing amplifier with T-resistor feedback path.

II. Proposed Amplifier Topology

A. Amplifiers Based on T-Resistor Feedback Network

Inserting a T-resistor network in the feedback path of a single-ended amplifier depicted in Fig. 2(a) is a known method [19] to increase the closed loop gain of an amplifier circuit requiring high input impedance. The signal gain and equivalent feedback resistance are defined in (5) and (6), respectively. By shunting some of the feedback signal via R_3 to ground, the effective feedback signal to the inverting terminal of the operational amplifier is reduced. To complete the negative feedback action, the OTA would need to generate a greater output signal to compensate for this feedback signal shunt loss, leading to a higher closed loop gain.

The T-network in the feedback path forms an equivalent feedback resistance defined by a composite sum of R_2, R_3 and $(R_1 \times R_2/R_3)$. As given in (6), this equivalent feedback resistance implemented by the T-network can achieve a much higher value without using any large resistances in the network. Thus,

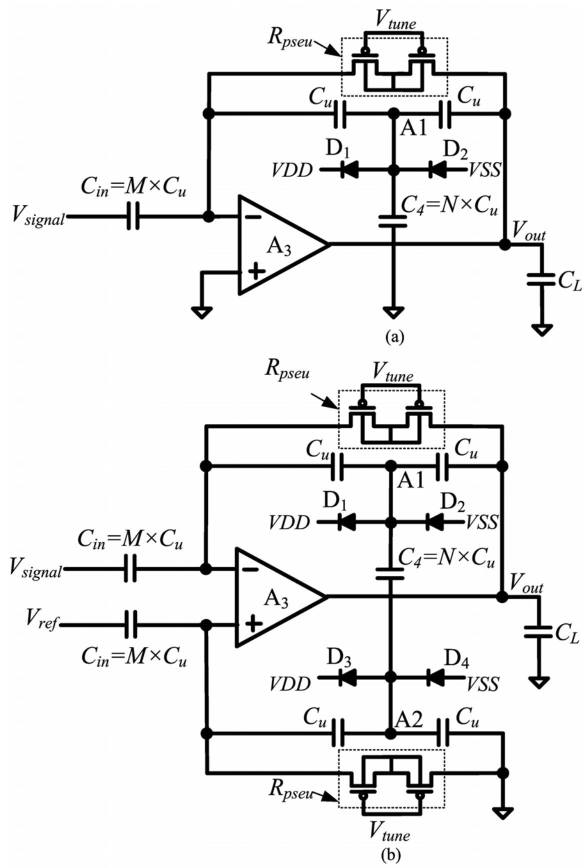

Fig. 3. Schematics of (a) TSCCA and (b) TDCCA.

high closed loop gain can be obtained while maintaining the sufficient input impedance.

A similar implementation for the differencing amplifier is illustrated in Fig. 2(b), and the effective feedback resistance is defined in (7).

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{R_{fb}}{R_{IN}} \tag{5}$$

$$R_{fbs} = R_1 + R_2 + \frac{R_1 R_2}{R_3} \tag{6}$$

$$R_{fbd} = R_1 + R_2 + \frac{2R_1 R_2}{R_3} \tag{7}$$

B. Extending to Capacitively Coupled Amplifiers

We extend the concept described in Section II-A to capacitively coupled amplifiers. Replacing the resistors in Fig. 2(a) with capacitors, a single-ended capacitively coupled amplifier with a T-capacitor feedback network (TSCCA) is obtained, as depicted in Fig. 3(a). To establish a proper DC bias, a pseudo-resistor is also added to the feedback path. This pseudo-resistor is implemented using a set of weakly inverted PMOS transistors. As defined in (2), this pseudo-resistor in conjunction with the effective feedback capacitor also determines the lower 3-dB cutoff frequency.

Similar to the T-resistor feedback path, the T-capacitor feedback network implements a smaller effective feedback capacitance given by

$$C_{fbeqs} = \frac{C_u}{(N+2)} \tag{8}$$

where N is the number of unit capacitors used to implement the shunt capacitor, C_4. This reduced equivalent feedback capacitance can be realized using conventionally-sized unit capacitors (C_u) that do not suffer from the problems associated with small size capacitors, as mentioned in Section I. As a result, an additional gain factor is obtained as compared with the conventional design in which only a single C_u is employed in the feedback path. Therefore to achieve the same mid-band gain, the input capacitance, C_{in} can be reduced by the same factor.

During the fabrication process or in the presence of external electric fields, parasitic charges may accumulate on the floating node A1 in Fig. 3(a). To eliminate this problem, reversely biased low leakage diodes (D_1 and D_2) are connected to the floating node. These diodes remain in reverse bias during normal operation and provide leakage paths to discharge any induced charge on the floating nodes, keeping the bias voltage on A1 within the safe operating voltage range. The diodes can be implemented with minimum size diffusions and hence have extremely low leakage current. The experiment results in Section IV show that these diodes have no significant impact on normal operation.

Similar to the TSCCA, the proposed differencing capacitively coupled amplifier (TDCCA) in Fig. 3(b) is derived from the circuit in Fig. 2(b). The smaller effective feedback capacitance is

$$C_{fbeqd} = \frac{C_u}{2(N+1)}. \tag{9}$$

Although more capacitors are used in the feedback path, the total number of unit capacitors needed to implement a given gain is in fact reduced and will be described further in Section II-C. Table I summarizes the key characteristics of the proposed topologies (TSCCA and TDCCA) versus those of conventional topologies (SCCA and DCCA).

C. Total Capacitance Reduction Versus Value of N

For each amplifier topology listed in Table I, the total number of unit capacitors required to implement a fixed gain is defined as $\mathrm{T_{CU}}$ and is a function of N and gain. Fig. 4 illustrates this function as $\mathrm{T_{CU}}$ is plotted versus N for all amplifiers implementing a fixed gain of 40 V/V. We can observe that for a reasonable value of N, the total number of unit capacitors needed to implement a given gain is smaller for TSCCA and TDCCA compared to the conventional topologies. For instance, to implement a differencing capacitively coupled amplifier with a gain of 40 V/V, the total number of C_u needed for the TDCCA is $16C_u$ (for $N = 4$) compared with $82C_u$ for the DCCA. This represents a 5.1 times reduction of the total capacitance. As the area of such capacitively coupled amplifiers is largely dependent on the number of capacitors in the circuit, substantial area can be saved with the TDCCA. On the other hand, if N is increased

TABLE I
KEY CHARACTERISTICS OF CAPACITIVELY COUPLED AMPLIFIERS. ALL CAPACITORS ARE IMPLEMENTED WITH UNIT CAPACITORS LABELED AS C_u

Topology	Fig.	Effective feedback capacitance (C_{fbeq})	Signal gain (G)	Higher 3 dB cutoff frequency (f_U)	Total unit caps to achieve gain, G (T_{CU})	Value of N for min. No. of unit caps (N_{min})
SCCA	1(a)	C_u	M	$g_m\left(\frac{1}{MC_L}\right)$	$G+1$	-
TSCCA	3(a)	$\frac{C_u}{(N+2)}$	$M(N+2)$	$g_m\left(\frac{1}{M(N+2)C_L}\right)$	$\frac{G}{N+2}+(N+2)$	$\sqrt{G}-2$
DCCA	1(b)	C_u	M	$g_m\left(\frac{1}{MC_L}\right)$	$2(G+1)$	-
TDCCA	3(b) and 4	$\frac{C_u}{2(N+1)}$	$2M(N+1)$	$g_m\left(\frac{1}{2M(N+1)C_L}\right)$	$\frac{G}{N+1}+(N+4)$	$\sqrt{G}-1$

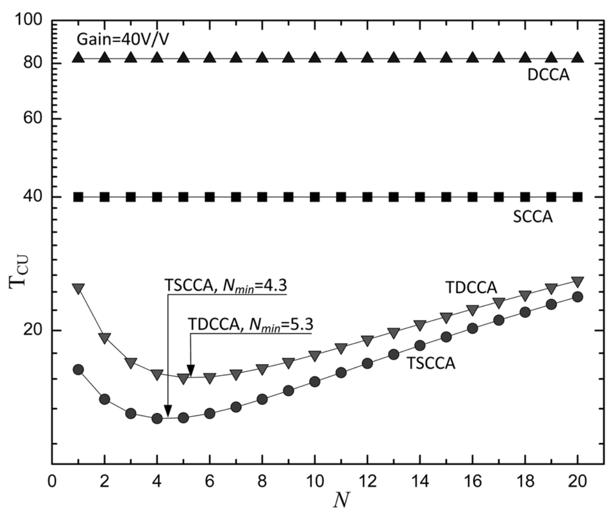

Fig. 4. Total number of unit caps to implement a gain of 40 V/V for various values of N.

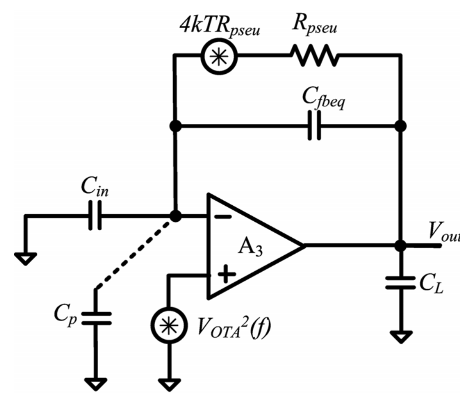

Fig. 5. Noise equivalent circuit for TSCCA, where C_p is the parasitic capacitance at input gate of OTA.

beyond $N_{\min}$, T_{CU} will start to increase. $N_{\min}$ (as defined in Table I) is the value of N that an amplifier can be implemented with a minimal number of unit capacitors for a given gain, G.

Although implementing the proposed amplifier with $N = N_{\min}$ capacitors will theoretically lead to the most compact area implementation, there will be more interconnections made between the feedback capacitors, as depicted in Fig. 3, leading to increased parasitic coupled capacitance between the nodes associated with the feedback interconnections. Such parasitic capacitors will cause the closed loop gain to deviate from the intended value. Increasing the value of N will also lead to increased input-referred noise as will be described next in Section II-D.

D. Noise Analysis

A noise analysis for the TSCCA in Fig. 3(a) can be made using the noise equivalent model of the TSCCA in Fig. 5. The input referred noise of the OTA is modeled as $V_{OTA}^2(f)$, and the pseudo-resistor is modeled by resistor R_{pseu} having noise power density of $4kTR_{pseu}$. The T-capacitor network is modeled with its equivalent capacitor C_{fbeq}, as defined in Table I, and the OTA input parasitic capacitor is modeled as C_p. The input referred noise power spectral density (PSD) as a function of frequency is analyzed, and we have

$$V_{IRF}^2(f) = \frac{4kTR_{pseu}}{f^2}\left(\frac{f_L}{G}\right)^2 + V_{OTA}^2(f)\left(\left(\frac{C_{in}+C_{fbeq}+C_p}{C_{in}}\right)^2 + \left(\frac{f_L}{G.f}\right)^2\right) \quad (10)$$

where G is the signal gain defined by the ratio C_{in}/C_{fbeq} and f_L is the lower 3 dB cutoff frequency of the recording amplifier defined by

$$f_L = \frac{1}{2\pi C_{fbeq}R_{pseu}}. \quad (11)$$

The parasitic capacitor, C_p, is predominantly contributed by the input gate capacitance of the OTA; ignoring layout-dependent parasitics, it can be simplified to

$$C_p = \alpha.W.L.C_{ox} \quad (12)$$

where α is a factor determined by the operating region of the transistors in the input differential pair. It is equal to 2/3 or $(1 -$

$1/\eta$) if the transistors operate in strong inversion or weak inversion regions, respectively. The factor, $1/\eta$ is the sub-threshold gate coupling coefficient. For a well optimized OTA, the input referred noise of the OTA is determined mainly by the noise of the input differential pair of the OTA [7], [20] and is given by

$$V_{OTA}^2(f) = \frac{16kT}{3g_m} + \frac{2K_F}{W.LC_{ox}f} \tag{13}$$

where C_{ox}, g_m, W and L are the unit gate oxide capacitance, transconductance, and the width and length of the input differential pair transistors, respectively. K_F is a process dependent constant.

The first term in (10) describes the low pass filtered thermal noise of the pseudo-resistor [21], which exhibits a brown noise power characteristic $(1/f^2)$. The second term is the input referred noise of the OTA $V_{OTA}^2(f)$ multiplied by a sum of 2 factors. The first factor describes the increase in noise due to the capacitive transformer formed by C_{in}, C_{fbeq} and C_p. The second factor is insignificant for frequencies above f_L/G and can be ignored. If the flicker noise component of $V_{OTA}^2(f)$ is low, the second term is approximately a flat power spectrum when referred to the input. The two noise components in (10) result in a corner frequency in the spectrum (when two noise densities are equal), which is found to be

$$f_c = f_L\left(\frac{C_{fbeq}}{C_{in}+C_{fbeq}+Cp}\right)\left(\sqrt{\frac{4kTR_{pseu}}{V_{OTA}^2(f_c)}}-1\right) \tag{14}$$

where $V_{OTA}^2(f_c)$ is the thermal noise floor of $V_{OTA}^2(f)$. (14) is valid if the flicker noise component of $V_{OTA}^2(f)$ has a corner frequency much lower than f_c.

Similar analyses can be performed for the TDCCA, and the input referred noise power spectral density and corner frequency of the TDCCA is found to be

$$V_{IRF}^2(f) = \frac{8kTR_{pseu}}{f^2}\left(\frac{f_L}{G}\right)^2 + V_{OTA}^2(f)\left(\left(\frac{(C_{in}+C_{fbeq}+C_p)}{C_{in}}\right)^2 + \left(\frac{f_L}{G.f}\right)^2\right) \tag{15}$$

and

$$f_c = f_L\left(\frac{C_{fbeq}}{C_{in}+C_{fbeq}+C_p}\right)\left(\sqrt{\frac{8kTR_{pseu}}{V_{OTA}^2(f_c)}}-1\right). \tag{16}$$

The definitions of f_L, C_p, and $V_{OTA}^2(f)$ remain the same, as defined in (11), (12) and (13) respectively.

Fig. 6 shows a plot of input referred noise PSD of the TSCCA as defined in (10) along with the magnitude response of the TSCCA. To maintain the same f_L when C_{fbeq} is reduced (by increasing *N*), R_{pseu} has to be increased by the same proportion, resulting in increased thermal noise density at low frequency and hence increased f_c (f_{c2}), as indicated in Fig. 6. On the other hand, for a very low f_L and small C_{fbeq}, the thermal noise due to R_{pseu} increases, but it concentrates at a lower frequency band because f_c is low.

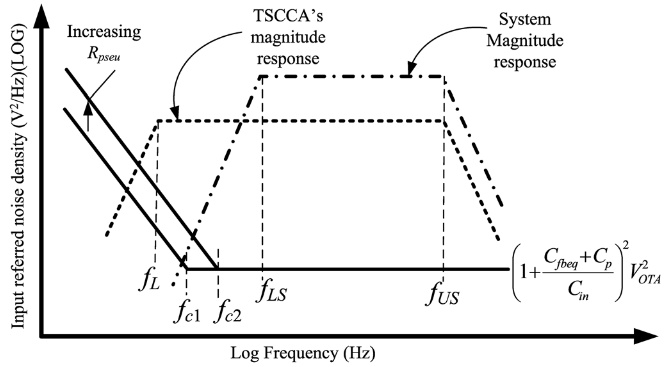

Fig. 6. Magnitude response and input referred noise density of TSCCA and recording system.

For a specific signal bandwidth (f_L to f_U) and ignoring the last factor in (10), the input referred mean squared noise power of the TSCCA can be obtained by substituting (12) and (13) into (10) and integrating (10) over the signal bandwidth. Hence, we obtain

$$\begin{aligned}\overline{V_{IN}^2} =& 4kTR_{pseu}\left(\frac{f_L}{G}\right)^2\left(\frac{1}{f_L}-\frac{1}{f_U}\right)\\ &+\left((f_U-f_L)\frac{16\pi kT}{6g_m}+\frac{2K_F}{W.LC_{ox}}\ln\left(\frac{f_U}{f_L}\right)\right)\\ &\cdot\left(\frac{C_{in}+C_{fbeq}+\alpha.W.L.C_{ox}}{C_{in}}\right)^2.\end{aligned} \tag{17}$$

Similarly, the input referred mean squared noise power of the TDCCA can be obtained as

$$\begin{aligned}\overline{V_{IN}^2} =& 8kTR_{pseu}\left(\frac{f_L}{G}\right)^2\left(\frac{1}{f_L}-\frac{1}{f_U}\right)\\ &+\left((f_U-f_L)\frac{16\pi kT}{6g_m}+\frac{2K_F}{W.LC_{ox}}\ln\left(\frac{f_U}{f_L}\right)\right)\\ &\cdot\left(\frac{C_{in}+C_{fbeq}+\alpha.W.L.C_{ox}}{C_{in}}\right)^2.\end{aligned} \tag{18}$$

In general for an amplifier system designed for neural spike recording, the lower 3-dB cut-off frequency is typically around 100 Hz. To reduce the total in-band noise, f_L should be set as low as possible (much lower than 100 Hz), yielding a low f_c. The system lower 3-dB cutoff frequency f_{LS}, as depicted in Fig. 6, can be defined by a subsequent signal conditioning stage [7], [22], [23]. In this way, the brown noise component (concentrated at low frequency band) can be significantly suppressed, and the overall input referred noise is independent of the value of C_{fbeq}. The first term in (17) and (18) then becomes negligible and both (17) and (18) can be reduced to

$$\begin{aligned}\overline{V_{IN}^2} =& \left(((f_{US}-f_{LS})\frac{16\pi kT}{6g_m}+\frac{2K_F}{W.LC_{ox}}\ln\left(\frac{f_{US}}{f_{LS}}\right)\right)\\ &\cdot\left(\frac{C_{in}+C_{fbeq}+\alpha.W.L.C_{ox}}{C_{in}}\right)^2.\end{aligned} \tag{19}$$

TABLE II
AMPLIFIER CONFIGURATIONS FOR INVESTIGATION

Amplifier	N	Gain (V/V)	C_{in} (pF)	C_{fbeq} (fF)
TDCCA1	2	78	2.6	33.33
TDCCA2	4	80	1.6	20.00
TDCCA3	6	84	1.2	14.28
DCCA1	-	80	16.0	200

It can be observed that at a particular $\mathrm{W.L} = (\mathrm{W.L})_{\min}$, the mean squared noise integrated over f_{US} and f_{LS} is minimal. This serves as a guideline for choosing the W and L of input transistors. If the operating bandwidth is within the region where $1/f$ noise is dominant, $(\mathrm{W.L})_{\mathrm{nmin}}$ would be higher. In other words, the minimal input referred noise in this case occurs at higher $(\mathrm{W.L})_{\mathrm{nmin}}$, compared to the case where thermal noise is dominant. This property will be fully illustrated in Section III-B.

As the device matching and therefore CMRR are directly proportional to the transistor gate area, an iterative approach to optimizing W.L versus noise and CMRR would have to be performed during the design.

To quantify the power-noise trade-off, the noise efficiency factor (NEF) [24], as defined in (20), is commonly adopted.

$$\mathrm{NEF} = \sqrt{\overline{V_{IN}^2}}\sqrt{\frac{2I_{tot}}{\pi.U_t.4\kappa T.BW}} \tag{20}$$

where I_{tot} is the total amplifier supply current, U_t is the thermal voltage at absolute temperature T and BW is the amplifier bandwidth $(f_{US} - f_{LS})$. Substituting (19) into (20), the final NEF of any amplifier can be obtained when accounting for thermal noise, flicker noise and the capacitive noise transformer effect. As the NEF is a direct function of the input referred mean squared noise power obtained in (19), it can be inferred that at the same $\mathrm{WL} = (\mathrm{WL})_{\mathrm{nmin}}$, the NEF would also be minimal.

III. NEURAL AMPLIFIER DESIGN

To demonstrate the robustness and feasibility of this proposed topology, TDCCA amplifiers with different values of N were designed. A conventional DCCA was also designed to provide a benchmark. Table I lists the types of configurations that were designed.

TDCCA1 is designed with $N = 2$ and has a higher input capacitance compared to the rest but according to (18) is expected to have lower input referred noise. TDCCA3 is designed with $N = 6$ to achieve a lower input capacitance but at the expense of higher input referred noise as predicted in (18). TDCCA2 is designed with $N = 4$ and is a good compromise between TDCCA1 and TDCCA3. According to Table I, $N = 8$ is needed to realize a TDCCA with a gain of 80 V/V using the minimum number of unit capacitors. However, it was found during the layout design that there is not much further reduction in area (as compared to TDCCA3) due to the layout overhead associated with the OTA and dummy capacitors.

Fig. 7(a) shows the pseudo-resistor design that is used for all the designed amplifiers. It occupies an insignificant on-chip area of 823 um^2. In general, for a given low side 3-dB cut-off frequency, the pseudo-resistor value and hence the transistor length need to be increased when C_{fbeq} is reduced. However, compared to the area occupied by the OTA and the unit capacitors, this area increase is insignificant.

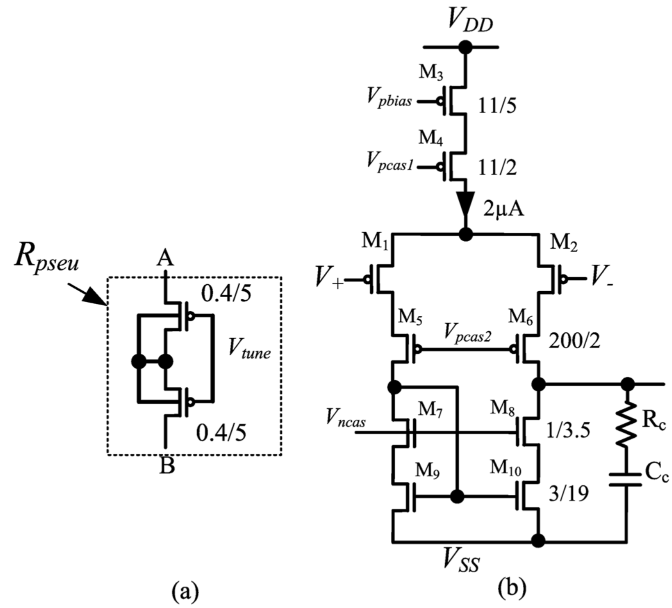

Fig. 7. (a). The pseudo-resistor used in all the proposed amplifiers. (b) Schematic of the OTA for the prototype amplifiers.

Comparing $\mathrm{TDCCA2}(N = 4)$ with the prior art having similar mid-band gain, around 6 to 12 times reduction in input capacitance is achieved without using very small unit capacitors in the feedback path.

A. OTA Design

The telescopic cascode OTA of Fig. 7(b) is selected as the active OTA for the prototype amplifiers as it has the lowest input referred noise level for a given bias current compared to other types of basic amplifiers [25]. Though it has a low output swing, it is suitable for the front end amplifier as the typical output swing after amplification is a maximum of $+/-40$ mV for a 500 μV neural signal. The W/L ratio of the input differential pair formed by M_1 and M_2 are sized to work in weak inversion and is optimized for low input referred noise as described next in Section III-B. Transistors $\mathrm{M}_5 - \mathrm{M}_{10}$ are designed to operate in strong inversion, and their noise contribution is less than 20% of the total input referred noise. The resistor $\mathrm{R_c}$ and capacitor $\mathrm{C_c}$ provide frequency compensation to allow the OTA to provide a phase margin of at least 60° for a capacitive load of 2 pF.

B. Noise and NEF Optimization

Based on (19) and (20), noise and NEF optimization of the designed amplifiers are performed by tuning the area (tuning L while W/L is fixed) of the input differential pairs over the 100 Hz–10 kHz signal bandwidth. The transistor width is maintained such that the input differential pair maintains a high $g_m/\mathrm{I_D}$ (>24) for all values of gate length. The target 0.35 μm process parameters were used with the Cadence Spectre simulator. For all amplifiers, their f_L is tuned to 0.1 mHz, which keeps the f_c of all amplifiers below 0.35 Hz. This decouples the brown noise contribution from the pseudo resistor from

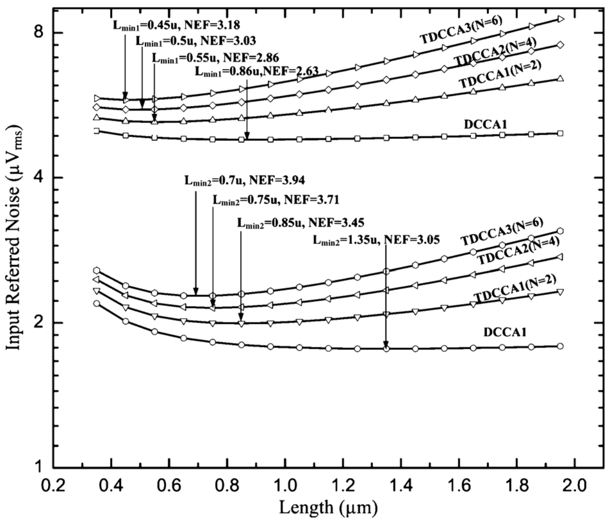

Fig. 8. Simulated noise optimization curves for different amplifiers. Upper 4 traces are input referred noise plots for 100 Hz to 10 kHz bandwidth. Lower 4 traces are input referred noise plots for 0.5 Hz to 1 kHz bandwidth. Width maintained at 150.5 × Length to keep g_m/I_D of M_1 and M_2 above 23 (weak inversion).

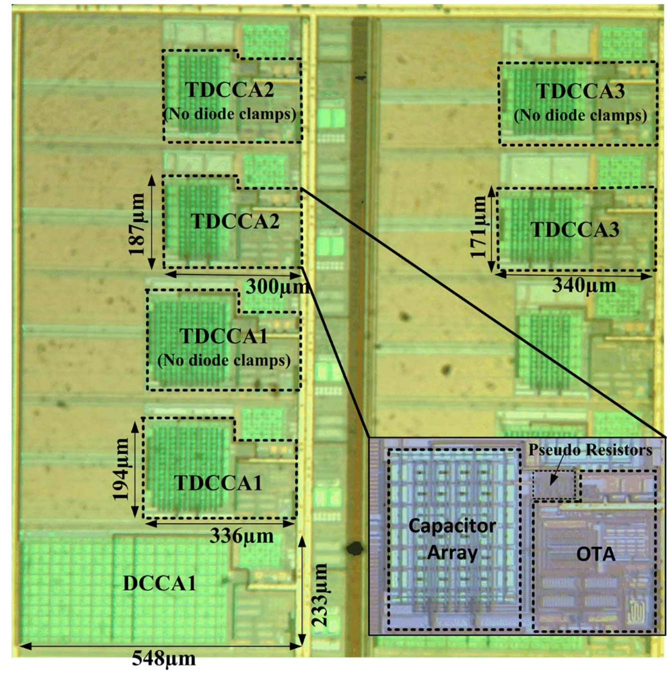

Fig. 9. Die Microphotograph of the neural amplifiers. A close up view of a TDCCA2 is also shown on the lower right of the micrograph.

TABLE III
FINAL OPERATING POINT FOR DEVICES IN OTA

Devices	M_1,M_2					
	W (μm)	L (μm)	g_m/I_D (V^{-1})	V_{gseff} (mV)	R_c (Ω)	C_c (pF)
TDCCA1	128	0.85	24.85	5.48	10	2.0
TDCCA2	105.6	0.7	24.70	5.59	18	1.36
TDCCA3	105.6	0.7	24.70	5.59	18	1.36
DCCA1	210.4	1.4	25.04	5.27	40	1.0

affecting the noise and NEF optimization. The upper 4 traces plotted in Fig. 8 show a plot of simulated RMS noise performance for the amplifiers versus input transistor gate length for signal bandwidth of 100 Hz–10 kHz. We observed that at a particular $\mathrm{L} = \mathrm{L}_{\min 1}$, the input referred noise and also NEF are minimal. To investigate the effect of having more flicker noise component in a noise optimization process based on (19), circuit simulations were also performed with a signal bandwidth of 0.5 Hz–1 kHz. Within this bandwidth, flicker noise is a dominant component of the OTA's input referred noise. The lower 4 traces in Fig. 8 show the simulated RMS noise performance for the designed amplifiers versus input transistor gate length for a signal bandwidth of 0.5 Hz–1 kHz. Similarly, we observed that at a particular $\mathrm{L} = \mathrm{L}_{\min 2}$, the input referred noise and NEF of each amplifier is minimal. Note that $\mathrm{L}_{\min 2}$ is higher than $\mathrm{L}_{\min 1}$, as predicted by (19) and discussed in Section II-D. Since the amplifiers are targeted for neural spike recording, the amplifiers were tuned for a signal bandwidth of 100 Hz–10 kHz. Taking process variations into consideration, the final gate lengths and widths of the input transistor pair of each amplifier are listed in Table III. The values of $\mathrm{R_c}$ and $\mathrm{C_c}$ for each amplifier are also included in this table.

IV. ELECTRICAL MEASUREMENT RESULTS

The prototype neural amplifiers (TDCCA1–3 and DCCA1) were fabricated, and the chip microphotograph of the amplifiers is shown in Fig. 10. The chip also includes TDCCA1–3 without the diode clamps (D1–D4) previously illustrated in Fig. 3(b). All capacitors were implemented with poly-poly capacitors having an area capacitance of 0.9 $\mathrm{fF}/\mu\mathrm{m}^2$. From Fig. 9, we can observe that the each of the proposed amplifiers (TDCCA1–3) occupies less area compared to the DCCA1.

For electrical characterization, the output of each neural amplifier was connected to an on-chip low-noise AC coupled buffer with 20 dB gain and 5 mHz lower 3-dB cutoff frequency. All parametric measures were taken from the output of these buffers. The f_L of each neural amplifier was tuned by supplying a voltage (V_{tune}) to the gates of the pseudo-resistors. This voltage was generated by an on-chip bandgap reference with an external tunable resistor.

Fig. 10 shows the AC frequency response of the TDCCA2 (with and without diode clamps) measured over 4 chips when f_L is tuned to 1 Hz. The widely varying f_L of the TDCCA2 is attributed to its pseudo-resistors whose values are process dependent. Fig. 11 shows the corresponding input referred noise root power spectral density (PSD) of the TDCCA2 and includes measurements on TDCCA2 without diode clamps. A close up view of the AC frequency responses and the input referred noise root PSD of TDCCA1–3 and DCCA1 in one of the chips are also shown in Fig. 12 and Fig. 13, respectively. We observed that the AC frequency responses and input referred noise PSD of any TDCCA with and without the diode clamps are very close. This shows that the diode clamps that were useful for preventing accumulated charges do not affect the performance of the proposed amplifiers. A less than 1.5 dB peaking at the low frequency end of the AC response for most TDCCAs was traced to the presence of parasitic capacitors at the middle nodes of the series connected pseudo-resistors. Such parasitic capacitors introduced an additional pole and zero around the low side 3-dB

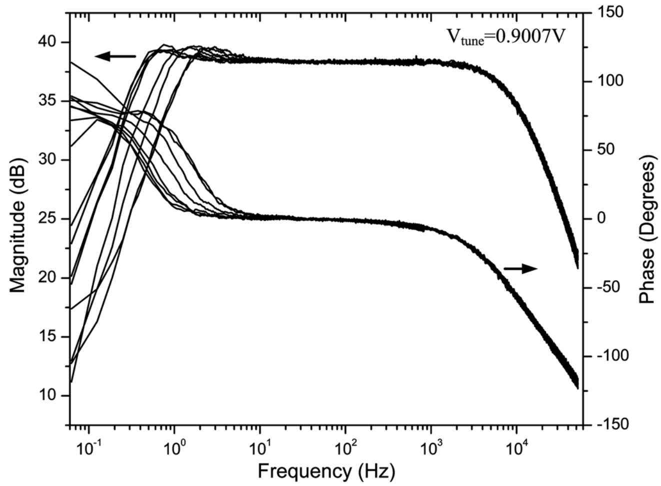

Fig. 10. AC response of the TDCCA2 (with and without diode clamps) measured over 4 chips.

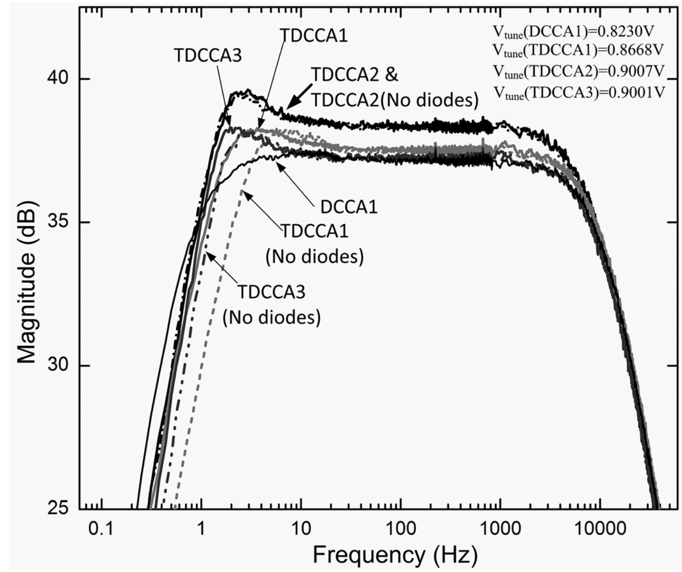

Fig. 12. Close up view of the low side AC response of TDCCA1–3 and DCCA1 of one the chips. TDCCA implementations without the diode clamps are also included.

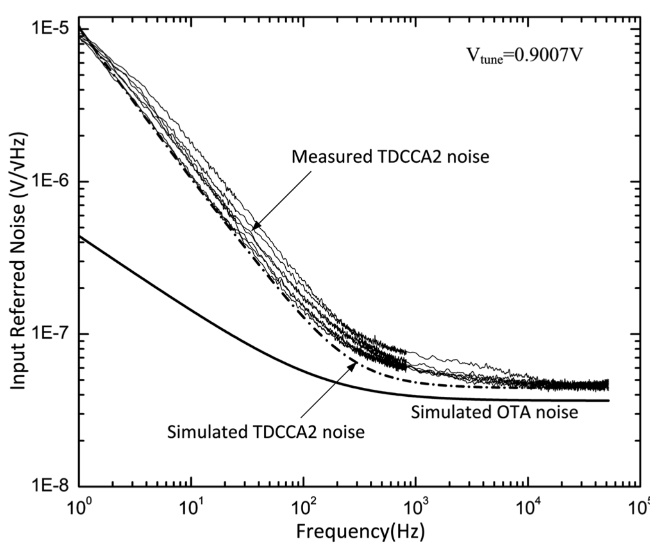

Fig. 11. Input referred noise PSD of the TDCCA2 (with and without diode clamps) measured over 4 chips. The simulated TDCC2 and its OTA input referred noise PSD are also included.

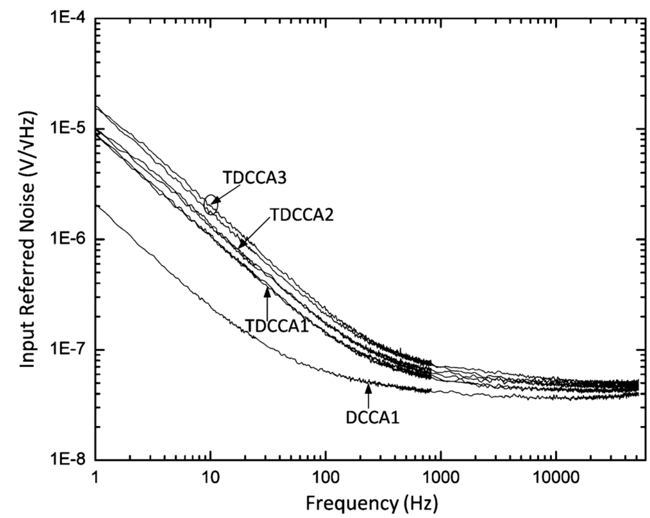

Fig. 13. Input referred noise PSD of the TDCCA1–3 (including those without diode clamps) and DCCA1 from a chip sample.

cutoff frequency leading to slight peaking at that frequency. The fabricated TDCCA2 achieved an average mid-band gain of 38.1 dB with 1.6 pF input capacitance (equivalent to an input resistance of 99 MΩ at 1 kHz), resulting in a C_{fbeq} of 20 fF, which is 5 times less than the state-of-art [11].

From Fig. 11, we observed that the measured input referred noise of the TDCCA2 is close to that of the simulated result. In addition, the simulated input referred noise of the OTA is also included, and its low frequency response shows a typical flicker noise type roll-off of $1/\sqrt{f}\ (\mathrm{V}/\sqrt{\mathrm{Hz}})$, compared to the TDCCA2 which has a typical $1/f\ (\mathrm{V}/\sqrt{\mathrm{Hz}})$ brown noise roll-off characteristic at low frequency. The brown noise described in Section II-D clearly dominates at low frequency for the TDCCA. When integrated over the operating bandwidth from 1 Hz–8.5 kHz, the RMS input referred noise of the TDCCA2 is 13.3 $\mu\mathrm{V_{rms}}$. The benchmark DCCA has only an RMS input referred noise of 4.88 $\mu\mathrm{V_{rms}}$ when integrated over the same bandwidth. The high input referred noise results from the thermal noise of the increased pseudo-resistance, as explained in Section II-D. For neural spike recording (100 Hz–10 kHz), most of the brown noise is out of band and can be removed by a high-pass filter having lower 3-dB cutoff frequency above 100 Hz. This high pass filter can be implemented in the subsequent low power stage similar to those in [7], [22], [23]. If such high-pass filter is added after the output of the TDCCA2, the total RMS input referred noise from 100 Hz to 8.5 kHz would be 6.07 $\mu\mathrm{V_{rms}}$.

From Fig. 13, it is observed that the input referred noise of all of the TDCCAs is higher than that of the DCCA. This is due to the increase in thermal noise resulting from the capacitive transformer effect as well as the increased brown noise defined in (18). For the f_L of 1 Hz, the brown noise level increases as N is increased. Similar to the analysis described earlier, a

TABLE IV
SUMMARY OF PERFORMANCE AND COMPARISON WITH THE STATE OF THE ART

Parameter	[10]	[11]	[17]	[26]	**This work (TDCCA1)**	**This work (TDCCA2)**	**This work (TDCCA3)**
CMOS Technology (μm)	0.18	0.35	0.35	0.13	**0.35**	**0.35**	**0.35**
Input load capacitance (pF)	20	4.5	5	30	**2.6**	**1.6**	**1.2**
Gain (dB)	39.4	33	34	47.5	**37.5**	**38.1**	**37.2**
C_{fbeq} (fF)	200	100	100	125	**33**	**20**	**14**
Operating Bandwidth (Hz)	10 -7.2k	0.5 -10k	0.01 -5k	11.5 -9.8k	**1 -10k**	**1 -8.5k**	**1-9k**
Amplifier Area (mm^2)	0.065**	0.02	0.02	0.062	**0.065**	**0.056**	**0.058**
Input Referred noise (μV_{rms})	3.5	6.08	7	3.8	**10.6**	**13.3**	**19.1**
Noise Bandwidth (Hz)	10 -7.2k	10 -5k	1 -5k	1 -6.9k	**1 -10k**	**1 -8.5k**	**1-9k**
NEF	3.35	5.55	4.6	2.16	**5.78**	**7.87**	**11.0**
Supply voltage (V)	1.8	3.0	3	1.2	**3**	**3**	**3**
Supply current (μA)	4.4	2.8	1.4	1.6	**2.0**	**2.0**	**2.0**
Power Consumption (μW)	7.92	8.40	4.2	1.92	**6.0**	**6.0**	**6.0**

** Effective area of a single amplifier as 4 amplifiers share a single OTA and a single reference input

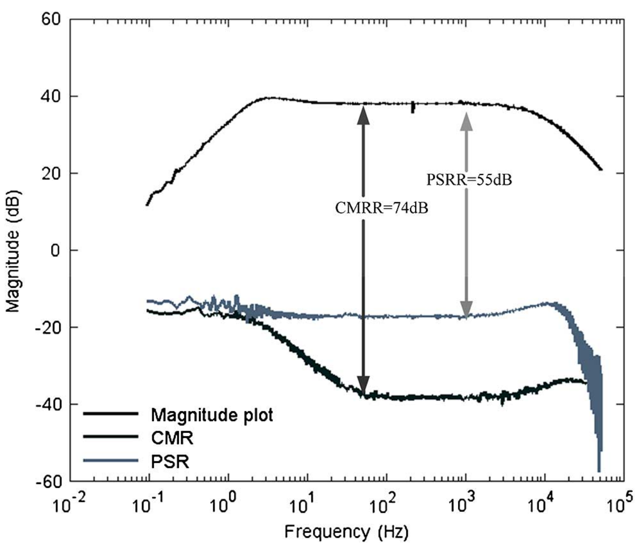

Fig. 14. CMRR and PSRR response of TDCCA2.

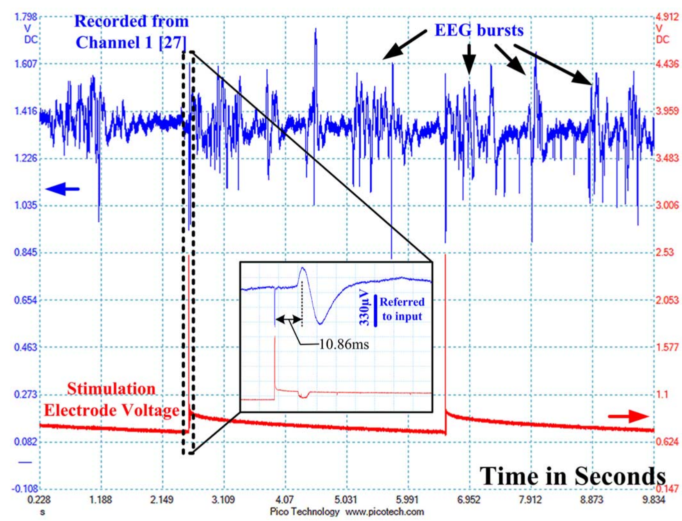

Fig. 15. Recorded SSEPs when the right fore limb of the rat is stimulated every 4 seconds.

high-pass filter would need to be added to remove the brown noise component for neural spike recording.

The measured CMRR and PSRR plots for the TDCCA2 are shown in Fig. 14. At 50 Hz, the neural amplifier's CMRR and PSRR are at least 74 dB and 55 dB, respectively. The total harmonic distortion at 1 kHz remains less than 1% for a maximum input peak to peak swing of 2.4 mV, and this is more than enough for the neural signals, which typically have input peak-to-peak signal swings of less than 2 mV.

Table IV summarizes the performance of the fabricated TDCCA neural amplifiers in comparison with the state of the art. All measured results in Table IV were obtained from first stage neural amplifiers. Measurement results for [11], [26] also include the common mode feedback circuit that are used to bias the fully differential neural amplifiers.

In amplifier systems having high channel counts [11], [17], gain was sacrificially reduced to achieve reduction in total chip area and lowered input loading capacitance. In [10], the effective area per amplifier was reduced by sharing the reference capacitor and the reference input branch of the OTA amongst 4 other signal input channels. It is possible to further share more channels with the reference circuitry but at the expense of increased CMRR, increased cross-talk, and increased timing constraints when the outputs are time-multiplexed. Nonetheless, the C_{fbeq} for the above mentioned designs was unable to fall below 100 fF and therefore limit further area reduction. On the contrary, our proposed work relaxed this constraint and achieved the lowest C_{fbeq} and the lowest input loading capacitance to date without compromising the other circuit performance parameters.

V. *In-Vivo* EXPERIMENT RESULTS

In addition to our *in-vivo* neural recording described in [18], *in-vivo* neural recordings have also been carried out using this amplifier at animal labs of the SINAPSE Institute, Singapore. All *in-vivo* experiments were performed in accordance with protocols approved by the Institutional Animal Care and Use Committee (IACUC) of the National University of Singapore.

The *in-vivo* neural recording was conducted using one Wistar rat that was prepared according to the surgical procedures described in [27] to allow chronic recording of somatosensory evoked potentials (SSEP). The cortical electrode placements are the same as defined in [27]. The rat was anesthetized with 2.5% isoflurane and its body temperature was maintained at 37°C. Sub-dermal needle electrodes were used to stimulate the median nerve of the right forelimb. Using a constant current stimulator (Digitimer DS3), unipolar current pulses of amplitude 3.5 mA with a duration of 200 μs were delivered once every 4 s to the electrodes on the right forelimb. Fig. 14 shows the output of the TDCCA2 amplifier recording SSEPs when the right forelimb was stimulated. From Fig. 14, we observed that whenever a stimulation current is delivered, the amplifier recorded a corresponding stimulation artifact followed by a typical SSEP [28]. The latency between the stimulation artifact and the first peak of the SSEP is 10.86 ms. Spontaneous EEG bursting was also recorded and can be observed between the stimulation pulses in Fig. 15.

VI. Conclusion

A neural recording amplifier with compact area and low input capacitance is proposed. By introducing a T-capacitor feedback network, the input capacitance (and area) versus gain constraint for the proposed amplifier is greatly relaxed. Single-ended and differencing amplifier topologies were discussed along with total capacitance and noise analysis of both types of amplifiers. Prototype neural amplifiers utilising the proposed topology for neural recording are demonstrated. Compared to the conventional capacitive coupled amplifier, the neural amplifiers described here achieved a low C_{fbeq} ratio without compromising other performance parameters. *In-vivo* recording experiments were also performed to demonstrate the capability of this amplifier for neural signal acquisition applications.

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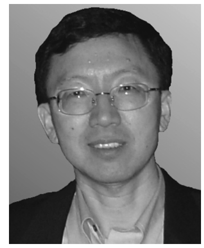

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